INCH-POUND

MIL-A-62565(AT)
22 June 1989
SUPERSEDING
(See 6.5)

MILITARY SPECIFICATION

AIR CLEANER, INTAKE: DRY-TYPE

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of automotive engine air cleaner consisting of a tubular inlet, a primary gravity precipitator with scavenging exhaust outlet, a replaceable filter element and a tubular outlet. The air cleaner is rated at 980 standard cubic feet per minute (scfm) [28 standard cubic meters per minute (m^3/min)] plus 10 percent (%) scavenging airflow (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2940

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SPECIFICATIONS

FEDERAL

P-C-437	- Cleaning Compound, High Pressure (Steam) Cleaner.
GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
QQ-P-416	- Plating, Cadmium (Electrodeposited).

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-P-19834	- Plates, Identification or Instruction, Metal Foil, Adhesive Backed, General Specification for.
MIL-C-22750	- Coating, Epoxy-Polyamide.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-45662	- Calibration Systems Requirements.
MS33660	- Tubing End, Hose Connection, Standard Dimensions for.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ARMY

12307204	- Air Cleaner, Intake.
12329100	- Filter Element.

(Copies of drawings required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B633	- Zinc on Iron and Steel, Electrodeposited Coatings of.
ASTM C800	- Glass Fiber Blanket Insulation (Aircraft Type).

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish air cleaners which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the unit to be furnished to the Government. All subsequent air cleaners delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Material shall be uniform in quality and free of defects which adversely affect performance or appearance of the finished product (see 4.7.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.3.1).

3.3 Design and construction. The air cleaner shall be fabricated as specified herein and in accordance with referenced specifications, standards and drawings. The air cleaner shall incorporate a primary gravity precipitator with scavenging exhaust outlet and a replaceable filter element (see 4.7.1 and 4.7.2).

3.3.1 Configuration. The air cleaner shall conform to Drawing 12307204 (see 4.7.1 and 4.7.2).

3.3.2 Dissimilar metals. The capability of dissimilar metal coupling shall be in accordance with MIL-STD-889 (see 4.7.1).

3.3.3 Standard parts. Military standard parts shall be used whenever possible (see 4.7.1).

3.3.4 Tubular joints. The tubular ends of the air cleaner inlet and outlet and the scavenging outlet shall have externally beaded ends conforming to MS33660, type A, except the bead height tolerance shall be plus (+) 0.003 inch [0.008 centimeter (cm)] and minus (-) 0.01 inch (-0.025 cm) (see 4.7.1 and 4.7.2).

3.3.5 Filter element. The filter element shall be of a one piece design, firmly bolted or clamped in place and easily replaced using hand clamps or common hand tools in accordance with Drawing 12329100 (see 4.7.2).

3.3.6 Interchangeability. All components having the same part number shall be dimensionally and functionally interchangeable (see 4.7.2).

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3.3.7 Weight. Weight of the air cleaner, including the filter element, shall not exceed 100 pounds (lb) [45.4 kilogram (kg)] (see 4.7.2).

3.3.8 Nameplate. The nameplate shall be as specified in either a or b (see 4.7.1 and 4.7.2).

a. Material and construction shall conform to MIL-P-514, type II, composition A, class 2, or composition C. Composition C material shall conform to GG-P-455, type II, grade A, class 1. Nameplate shall be attached with suitable metal fasteners to achieve a permanent installation.

b. Adhesive-backed aluminum foil shall conform to MIL-P-19834, type II, glossy, except backing sheet, which may be of paper, shall be removable intact by stripping without use of water or other solvents.

3.3.9 Safety. The air cleaner shall be designed so that safety hazards to personnel or equipment are minimized (see 4.7.1).

3.3.10 Flame resistance. After the filter media of the filter element is exposed to a flame until it burns or glows, the filter media shall not continue to burn or smolder for more than 10 seconds when the flame is removed (see 4.7.3).

3.4 Performance. Except as otherwise specified herein, performance requirements shall be met under the following conditions (see 4.7.4):

a.	Temperature	-80 ± 9° Fahrenheit (°F) [27 ± 5 degrees Celsius (°C)].
b.	Humidity	- 35 ± 5%.
c.	Density of air	-.073 pound per cubic foot (lb/ft^3) at 80°F and 29.92 inches of mercury (Hg) [1.2 kilogram per cubic meters (kg/m^3) at 27°C and 760 millimeters of mercury.]
d.	Pressure	- 760 ± 25 millimeters of mercury.
e.	Dust	- Chemical analysis of the test dust shall be as shown in table I. Particle size distribution shall be as shown in table II.
f.	Dust feed rate	- 0.025 gram/cubic ft [0.883 ± 0.176 gram per cubic meters (g/m^3).]

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TABLE I. Chemical analysis of test dust.

Substance	Percent by weight
Silicon dioxide (SiO_2)	67 to 69
Ferric oxide (Fe_2O_3)	3 to 5
Aluminum oxide (Al_2O_3)	15 to 17
Calcium oxide (CaO)	2 to 4
Magnesium oxide (MgO)	0.5 to 1.5
Total alkalis	3 to 5
Ignition loss	2 to 3

TABLE II. Particle size distribution.

Particle size (microns)	Percent of total weight	
	Coarse grade	Fine grade
0 - 5	12 ± 2	39 ± 2
5 - 10	12 ± 3	18 ± 3
10 - 20	14 ± 3	16 ± 3
20 - 40	23 ± 3	18 ± 3
40 - 80	30 ± 3	9 ± 3
80 - 200	9 ± 3	---

3.4.1 Air cleaning capacity. The air cleaner shall have an air flow capacity of 1078 scfm (31 m^3/min.) supplied to the precleaner. 980 scfm (28 m^3/min.) is the clean air flow through the filter element to the engine and the 10% scavenging flow approximate the required 1078 scfm (31 m^3/min.) air cleaning capacity (see 4.7.4.1).

3.4.2 Air leakage resistance. The air cleaner, when subjected to a vacuum of 50 inches of water or a pressure of 2 pounds per square inch (psi) [14 kilopascals (kPa)] for a period of not less than 5 minutes shall not leak more than 0.3 cfm (0.008 m^3/min) (see 4.7.4.2).

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3.4.3 Initial airflow restriction. Pressure drop from intake to outlet shall not exceed 9.6 inches of water (2.4 kPa) when the air filter is clean and airflow is 1078 scfm (980 scfm + 10% scavenging flow) [31 m^3/min (29 standard m^3/min + 10% scavenging flow)] (see 4.7.4.3).

3.4.4 Efficiency. The air cleaner efficiency shall be not less than 99.5% when subjected to an airflow of 1078 scfm (980 scfm + 10% scavenging flow) [31 m^3/min (28 standard m^3/min + 10% scavenging flow)] and fed fine dust in accordance with table II for 60 minutes (see 4.7.4.4).

3.4.5 Service life. When operating to the stepped airflow cycle of Figure 6, dust feed rate of paragraph 3.4f and scavenging flow rates at 10 percent of each airflow percent cycle run, the dust capacity of the air cleaner shall not exceed 20 inches of water (5 kPa) after 28 hours of operation (see 4.7.4.5).

3.4.6 Post service airflow restriction. Pressure drop across the air cleaner shall not exceed 1 inch of water (0.25 kPa) over original pressure drop after the air cleaner has been operated to dust capacity, cleaned and the same filter element reinstalled (see 4.7.4.6).

3.5 Environmental.

3.5.1 Shock. The air cleaner shall meet the performance requirements of 3.4.1, 3.4.2 and 3.4.4 after being subjected to a sawtooth shock pulses of 40 gravity units for 11 milliseconds (ms) time duration applied along each of the three major axes (see 4.7.5.1).

3.5.2 Vibration. The air cleaner shall meet the performance requirements of 3.4.1, 3.4.2 and 3.4.4 after exposure to vibration levels of 1.5 and 4.2 gravity units in the frequency range of 5 to 500 to 5 hertz (Hz) with a sweep time of 15 minutes (see figure 1 and 4.7.5.2).

3.5.3 Steam and waterjet cleaning. The air cleaner shall meet the performance requirements of 3.4.1, 3.4.2 and 3.4.4 after being subjected to high pressure steam cleaning using cleaner conforming to P-C-437 and then a waterjet cleaning. The jet shall be applied perpendicular to the air cleaner surface at a distance of not more than 12 inches (30 cm) for steam cleaning and not more than 60 inches (153 cm) for waterjet cleaning. The jet pressure for both water and steam shall be 110 to 110 psig (700 to 760 kPa) and the jet shall be moved at such a rate that the exposed surface of the air cleaner is subjected to the jet at a rate of 1 square foot/minute [0.1 square meter per minute (m^2/min)] (see 4.7.5.3).

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3.6 Finish (see 4.7.2).

3.6.1 Plating. The filter element carrier and latch shall be cadmium plated in accordance with QQ-P-416, type II, class 3 or zinc plated in accordance with ASTM B633, type II, service conditions 2 (SC-2). The zinc plating shall be not less than 0.3 inch (7.6 mm) thick.

3.6.2 Coating. The exterior surfaces of the air cleaner shall be coated in accordance with MIL-C-5541, class 1A.

3.6.3 Painting. Primer and topcoat shall be applied to the exterior surfaces of the air cleaner case only.

3.6.3.1 Primer. The exterior surfaces of the air cleaner case shall be primed in accordance with MIL-P-53030 or MIL-P-53022 or MIL-P-23377, composition optional. The dry film thickness of the undercoat shall be not less than 1.0 mil.

3.6.3.2 Topcoat. The exterior surfaces of the air cleaner case shall be topcoated with an epoxy-polyamide coating conforming to type I of MIL-C-22750 or a modified high solids epoxy which meets the performance requirements of MIL-C-22750. The dry film thickness of the topcoat shall be not less than 1.0 mil. The color shall match color chip number 24533 (green) of FED-STD-595.

3.7 Identification marking. Unless otherwise specified (see 6.2), the air cleaner shall be identified in accordance with MIL-STD-130 (see 4.7.2).

3.8 Workmanship. Workmanship shall be of such quality as to assure that the air cleaner shall be free from defects that reduce the capability of the air cleaner in the performance of the intended use or affects its appearance. The air cleaner shall be free of defects such as burrs, scratches, chips, sharp edges, corrosion, scale and dirt (see 4.7.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10% of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.2.1 Special test equipment.

4.1.2.1.1 Setup for tests requiring airflow. The test setup for all tests requiring airflow shall be as specified in figure 2.

4.1.2.1.1.1 Piezometer tubes, transition ducts, adapter, manometer and flow meter. Piezometer tubes conforming to figure 3 shall be installed at the inlet and outlet of the air cleaner. The inside diameter of the piezometer tubes shall be equal to the inside diameter of the air cleaner outlet (+ 0.5, - 0.0 inches) (+ 1.27, - 0.0 cm). Transition ducts shall be installed, if required, between the air cleaner inlet and inlet piezometer tube. Transition ducts and adapters shall have sides at angles no greater than 7 degrees measured from the centerline of the duct or adapter. A smooth approach (see figure 4) shall be installed at the front of the inlet piezometer tube. A water manometer shall be installed between the two piezometer tubes (see figure 2). A flowmeter shall be connected to the outlet of the master filter.

4.1.2.1.1.2 Master filter. The master filter shall consist of one layer of 0.5 inches (1.27 cm) fiberglass batting conforming to ASTM C800 installed in a suitable housing with the nap side of fiberglass batting facing upstream. The master filter batting shall be of sufficient size to prevent excessive restriction. The face velocity of air against the master filter batting shall not exceed 200 feet (60 meters) per minute. A manometer shall be located in the test setup to measure the pressure drop across the master filter (see figure 2). A new master filter batting shall be installed if, during the dust capacity test (see 3.4.5 and 4.7.4.5), the air pressure drop across the master filter increases by 10 inches of water (2.5 kPa) above its original restriction. The filter housing shall be electrically grounded during tests.

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4.1.2.1.1.2.1 Preparation of master filter. The fiberglass batting shall be hung in an air oven and dried for 6 hours at a temperature of 200 to 240°F (93 to 115°C). Drying time may be reduced, and the batting considered completely dry, if there is no change in the weight of the batting in two consecutive 10 minutes intervals. All weighings are to be done with the fiberglass batting in the oven. The drying and oven-weighing procedure shall be observed before and after each test involving dust feeding.

4.1.2.1.1.3 Scavenging vacuum. During all tests, except dust capacity test, a scavenge vacuum of 5 inches of water (1.25 kPa) or continuous air flow of 98 cfm (2.8 m^3/min) shall be applied to the scavenge outlet. A piezometer tube of suitable capacity, a manometer and a dust collection system shall be connected to the scavenge outlet (see figure 2).

4.1.2.1.1.4 Dust feeder. The dust feeder (see figure 5) shall be of a type that will deliver dust to the cleaner without breaking down the dust particles or creating conglomerates. The feeder shall deliver the dust uniformly throughout the test. Dust at the beginning of testing shall have the same particle distribution as dust at the conclusion of testing.

4.2 Classification of inspections:

a. First article inspection (see 4.4).

b. Quality conformance inspection (QCI) (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature: 73 ± 18°F (23 ± 10°C)
b. Barometric pressure: $28.5 \begin{smallmatrix}+2\\-3\end{smallmatrix}$ inches of mercury
   [$725 \begin{smallmatrix}+50\\-75\end{smallmatrix}$ millimeter of mercury]
c. Relative humidity: 50 ± 30%
d. Performance tests shall be conducted under the conditions specified in 3.4.

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select three air cleaners produced under the production contract for first article inspection. First article samples shall be inspected as specified in table III. Approval of the first article sample by

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the Government shall not relieve the contractor of his obligation to supply air cleaners that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance			
				Examination	Tests		Control
					Group A	Group B	
Materials and construction	3.2 thru 3.3.4, 3.3.8 and 3.3.9	4.7.1	X				
Defects (see table IV)	3.3, 3.3.1, 3.3.4 thru 3.3.8 and 3.6 thru 3.8	4.7.2	X	X			
Flame resistance	3.3.10	4.7.3	X				
Air cleaning capacity	3.4.1	4.7.4.1	X				
Air leakage resistance	3.4.2	4.7.4.2	X		X		
Initial airflow restriction	3.4.3	4.7.4.3	X				X
Efficiency	3.4.4	4.7.4.4	X			X	
Service Life	3.4.5	4.7.4.5	X				X
Post service airflow restriction	3.4.6	4.7.4.6	X				X
Shock	3.5.1	4.7.5.1	X				
Vibration	3.5.2	4.7.5.2	X				
Steam and water jet cleaning	3.5.3	4.7.5.3	X				

4.4.1 First article inspection failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of, first article inspection shall be evidence that all items already produced prior to completion of the first article inspection are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

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4.5 QCI.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the air cleaners from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sampling for group B tests. Samples for group B tests shall be selected in accordance with general inspection level II of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level (AQL). Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following AQL's:

Classification	AQL
Major	2.5
Minor	4.0

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table IV.

TABLE IV. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major	AQL 2.5% Defective	
101	Nonconformance in design and construction (see 3.3, 3.3.1, 3.3.4 thru 3.3.6).	Visual and SIE 1/
102	Dimensions affecting interchangeability, not within tolerance (see 3.3.1 and 3.3.6).	SIE
103	Faulty workmanship affecting performance (see 3.8).	Visual

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TABLE IV. Classification of defects - Continued.

Category	Defect	Method of examination
Minor	AQL 4.0% Defective	
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3.1 and 3.3.6).	SIE
202	Weight, not as specified (see 3.3.7).	SIE
203	Nameplates, improper (see 3.3.8).	Visual
204	Finish, improper (see 3.6).	Visual
205	Identification marking, improper (see 3.7).	Visual
206	Faulty workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Tests.

4.5.3.1 Group A. Each air cleaner shall be subjected to the test specified in table III.

4.5.3.1.1 Failure. Failure of any air cleaner to pass the test shall be cause for the Government to refuse acceptance of that air cleaner.

4.5.3.2 Group B. Samples selected in accordance with 4.5.1.2 shall be subjected to the tests specified in table III, and shall conform to an AQL of 2.5.

4.6 Control tests. Control tests shall be conducted on 1 air cleaner from each lot of 50 units consecutively produced. Samples shall be subjected to the tests specified in table III.

4.6.1 Control test failure. If the sample air cleaner fails to pass any of the control tests, two additional air cleaners shall be subjected to control tests. Failure of either air cleaner to pass any of the specified control test shall be cause for the Government to refuse acceptance of the production quantity represented, until corrective action taken by the contractor is acceptable to the Government.

4.6.2 Disposition of sample units. Air cleaners which have been subjected to control testing may be delivered on the contract.

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4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2 through 3.3.4, 3.3.8 and 3.3.9 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.3.1, 3.3.4 through 3.3.8 and 3.6 through 3.8 shall be determined by examination for the defects listed in table IV. Examination shall be visual or by measurement with standard inspection equipment.

4.7.3 Flame resistance. To determine conformance to 3.3.10, expose a sample of the filter media used in the filter element to a flame until the sample burns or glows. Remove the flame, observe and record results.

4.7.4 Performance. Except as otherwise specified herein, test conditions shall be in accordance with 3.4 and the following:

a. The test setup for all tests requiring airflow shall be as specified in 4.1.2.1.1.
b. Dust feed rates shall be averaged over 10-minute intervals.
c. Measured airflow restriction values shall be corrected to the specified air density (see 3.4c) to an accuracy of 0.1 inch of water (.025 kPa).
d. Actual airflow values shall be corrected to standard conditions (see 3.4c) and shall be within + 2% of specified values.
e. The assembly shall be mounted in its normal operating attitude and shall be electrically grounded.
f. For first article testing and control testing two filter elements shall be used for each air cleaner tested. The second filter element shall be inserted prior to the dust capacity test.
g. The air cleaner shall be stabilized at ambient conditions 1 hour before testing.

4.7.4.1 Air cleaning capacity. To determine conformance to 3.4.1, the successful completion of test procedures in 4.7.4.3 through 4.7.4.6 where airflow is adjusted to, 1078 scfm (31 standard m^3/min) (including scavenging flow) shall verify the air cleaner capacity requirement.

4.7.4.2 Air leakage resistance. To determine conformance to 3.4.2, the air cleaner inlet shall be connected to a flowmeter and all other inlets and outlets shall be sealed to prevent air leakage. The flowmeter shall be as accurate as a .187 inch (4.76 millimeters) orifice connected to a ± .01 inch (± .025 cm) water manometer. The flowmeter shall indicate an airflow no greater than 0.3 cfm (0.0085 m^3/min) when the air cleaner is subjected to a vacuum of 50 inches of water (14 kPa) when measured at the air cleaner air

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outlet. As an alternate method, the inlet and outlet of the cleaner shall be sealed. The interior of the cleaner shall be pressurized at no less than 2 psi (14 kPa) air pressure through a pipe coupling inserted in a drilled and tapped hole in the inlet/outlet sealing plate. There shall be no evidence of air leakage in the flow of bubbles escaping from the air cleaner when the air cleaner is submerged in water.

4.7.4.3 Initial airflow restriction. To determine conformance to 3.4.3, the following test shall be conducted. With the air cleaner set up as shown in figure 2, airflow restriction in inches of water shall be determined and recorded at the rated airflow of 1078 scfm (980 scfm + 10% scavenging flow) [31 m^3/min (28 standard m^3/min + 10% scavenging flow)].

4.7.4.4 Efficiency. To determine conformance to 3.4.4, the following test shall be conducted. With the air cleaner set up as shown in figure 2, the airflow shall be adjusted to 1078 scfm (980 scfm + 10% scavenging flow) [31 m^3/min (28 standard m^3/min + 10% scavenging flow)]. Fine dust shall be fed in accordance with 3.4f for a period of 60 minutes with airflow restriction recorded at start of test; and at 10 minute intervals. The master filter shall be dried and oven-weighed in accordance with 4.1.2.1.1.2.1 taking care to retain all entrapped dust. Weighings shall be accurate to 0.01 gram, wherever possible, but never less accurate than 0.1 gram. Efficiency shall be determined by means of the following equation:

$$\text{Efficiency \%} = \frac{W_1 - (W_2 - W_3)}{W_1} \times 100$$

Where: W_1 is the weight of dust fed into air cleaner inlet.

W_2 is the weight of master filter and entrapped dust.

W_3 is the weight of master filter before tests.

4.7.4.5 Service life. To determine conformance to 3.4.5, the following dust capacity test shall be conducted. A new filter element shall be installed for this test. With the air cleaner set up as shown in figure 2, the airflow shall be adjusted to 1078 scfm (980 scfm + 10% scavenging flow) [31 m^3/min [28 standard m^3/min + 10% scavenging flow (2.8 m^3/min)]. The airflow restriction shall be recorded at the start of testing and at intervals not to exceed 60 minutes throughout the test. Coarse dust shall be fed in accordance with 3.4f. After obtaining the initial airflow restriction at the rated airflow, the airflow shall be varied as shown in figure 6, with each interval of constant flow being maintained for approximately 10 minutes. Scavenging flow shall be maintained at 10% of each airflow percent cycle run (for example 80, 60, 40, and 20). This airflow cycle shall be continued for the duration of this test. The dust capacity test shall be terminated when the airflow restriction exceeds 20 inches (51 cm) of water, or at the end of the 28 hours, whichever occurs first. Prior to removal of the filter element from the air cleaner, carefully observe the clean side of the filter element for visual signs of dust tracking on the clean side of the seal or the filter media. Any visual sign of dust tracking on the clean side of the filter media or seal shall be a cause for rejection.

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4.7.4.6 Post service airflow restriction. To determine conformance to 3.4.6, after the dust capacity test (see 4.7.4.5), the filter element shall be removed from the air cleaner and compressed air cleaned as follows: Insert the air hose nozzle into each element pocket and blow the deposited dust from the filter element using a back and forth motion. The element must be first blown in a direction opposite to normal air flow and then in a direction normal to air flow. Special attention must be given to removing dust from the edges and seams. Line air pressure shall be 100 ± 10 psi (690 ± 70 kPa) and the nozzle inside diameter shall be a nominal .125 inch (.318 cm.). After the filter element is thoroughly cleaned it shall be reassembled in the air cleaner. Air flow shall be adjusted to 1078 scfm (980 scfm + 10% scavenging flow) [31 m^3/min (28 standard m^3/min + 10% scavenging flow)] and the airflow restriction determined.

4.7.5 Environmental.

4.7.5.1 Shock. To determine conformance to 3.5.1, the air cleaner shall be mounted in a position simulating its actual use. Three shocks in each direction shall be applied along the 45 inches (114 cm) dimension, 38 inches (96.5 cm) dimension and 10 inches (25.4 cm) dimension (total of 18 shocks). The shock pulse shape shall be in accordance with figure 7, of amplitude 40 gravity units and 11.0 ms time duration. At the conclusion of the test, the air cleaner shall pass the tests specified in 4.7.4.1, 4.7.4.2 and 4.7.4.4.

4.7.5.2 Vibration. To determine conformance to 3.5.2, the air cleaner shall be attached by its normal mounting means, either directly to the vibration excitor or transition table or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. The air cleaner shall be vibrated along each axis in accordance with the test level specified in figure 1. When the air cleaner resonances below 5 Hz are measured, the test curve shall be extended to 2 Hz (2 to 500 to 2 Hz) and the sweep time shall be increased by 3 minutes (a total of 18 minutes). The frequency of applied vibration shall be swept over the specified range logarithmically in accordance with figure 8. The specified sweep time is that of an ascending plus a descending sweep shown and is twice the ascending sweep time shown on figure 8. The sinusoidal cycling times shall be 30 minutes per axis. At the conclusion of the test, the air cleaner shall pass the tests specified in 4.7.4.1, 4.7.4.2 and 4.7.4.4.

4.7.5.3 Steam and waterjet cleaning. To determine conformance to 3.5.3, the air cleaner shall be subjected to the cleaning specified in 3.5.3. Subsequently the air cleaner shall pass the tests specified in 4.7.4.1, 4.7.4.2 and 4.7.4.4.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging requirements specified by the contracting authority (see 6.2).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The air cleaner covered by this specification is intended for use on military tracked vehicles in the M2/M3 Bradley Fighting Vehicle Systems (FVS).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If first article inspection is not required (see 3.1).
c. If identification marking shall be other than as specified (see 3.7).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. If first article inspection sample size shall be other than as specified (see 4.4).
h. Selection of applicable level and packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.3.2).

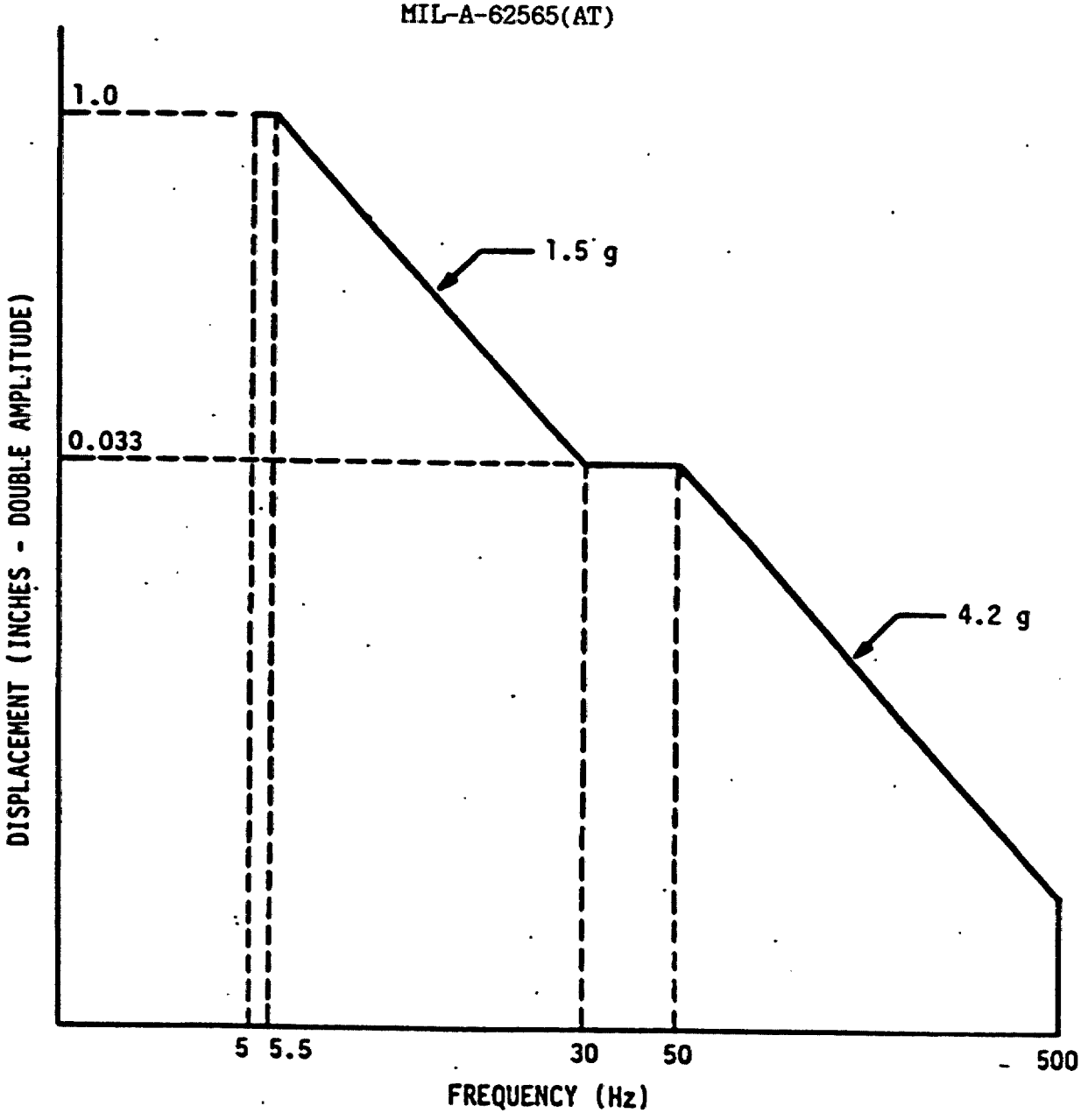
6.3.2 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.4 Subject term (key word) listing.

Air cleaning capacity
Efficiency
Filter element

6.5 Supersession data. This military specification supersedes FMC Corporation specification 12292158, revision D, dated 4 November 1986.

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NOTES:

1. Acceleration levels: ± g.
2. The curve shall be extended to 2 Hz when test item resonances below 5 Hz are expected.

FIGURE 1. Vibration test curves for equipment installed in ground vehicles.

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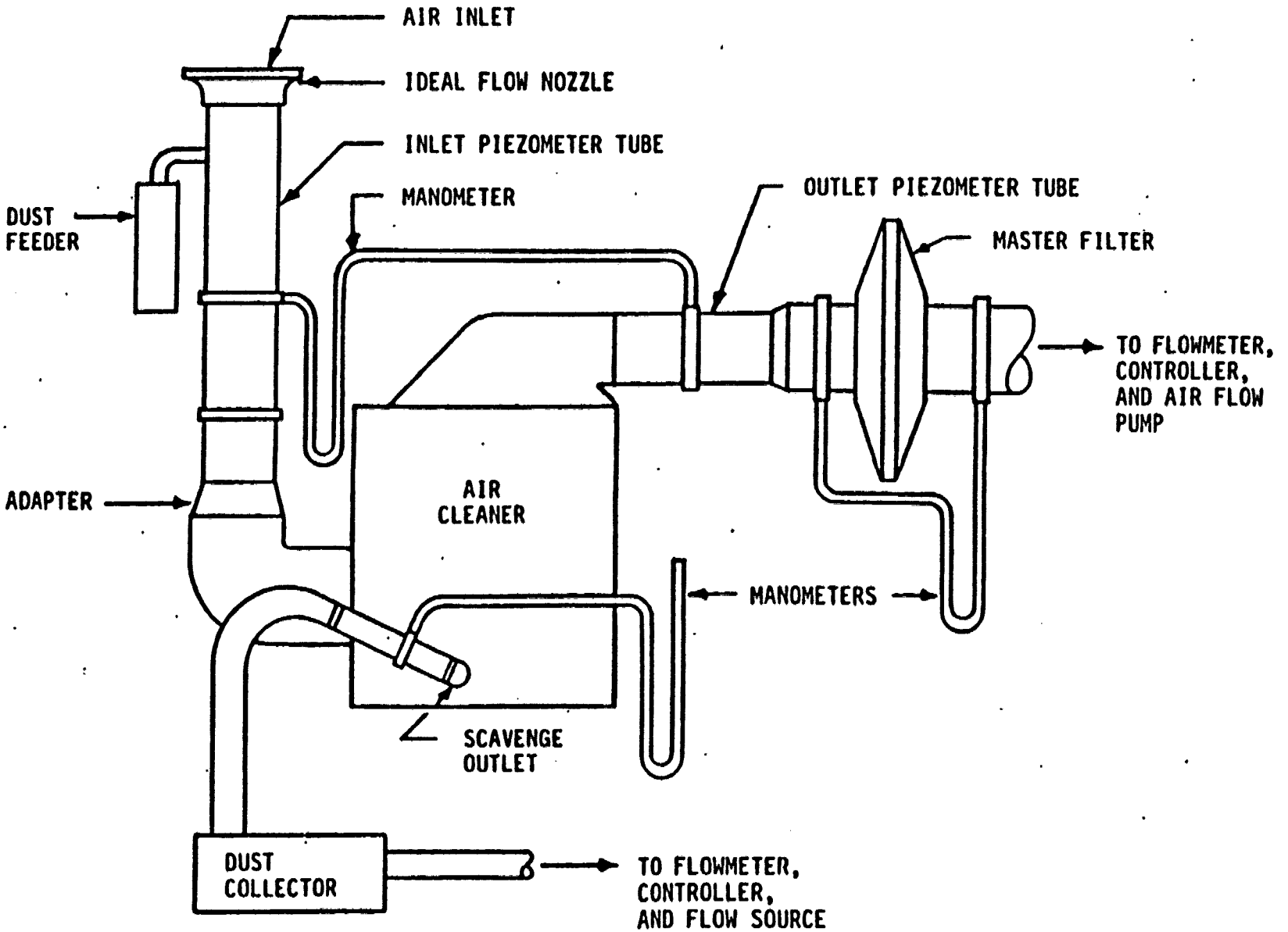
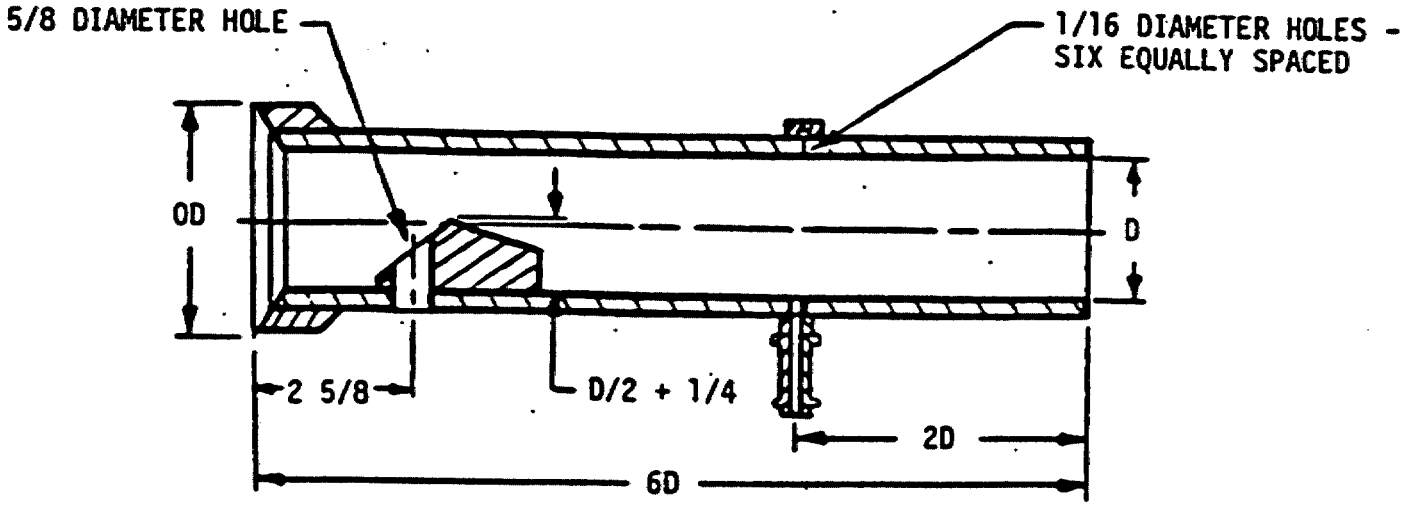
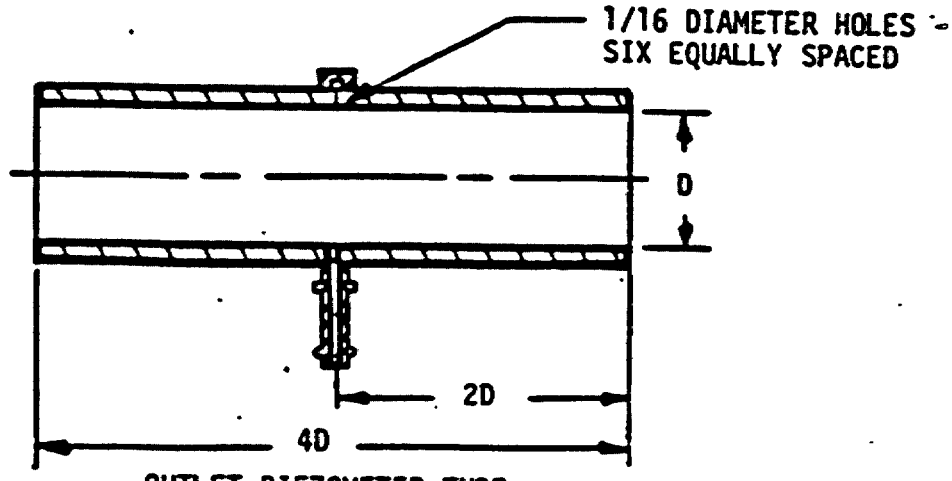


FIGURE 2. Test setup.

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INLET PIEZOMETER TUBE



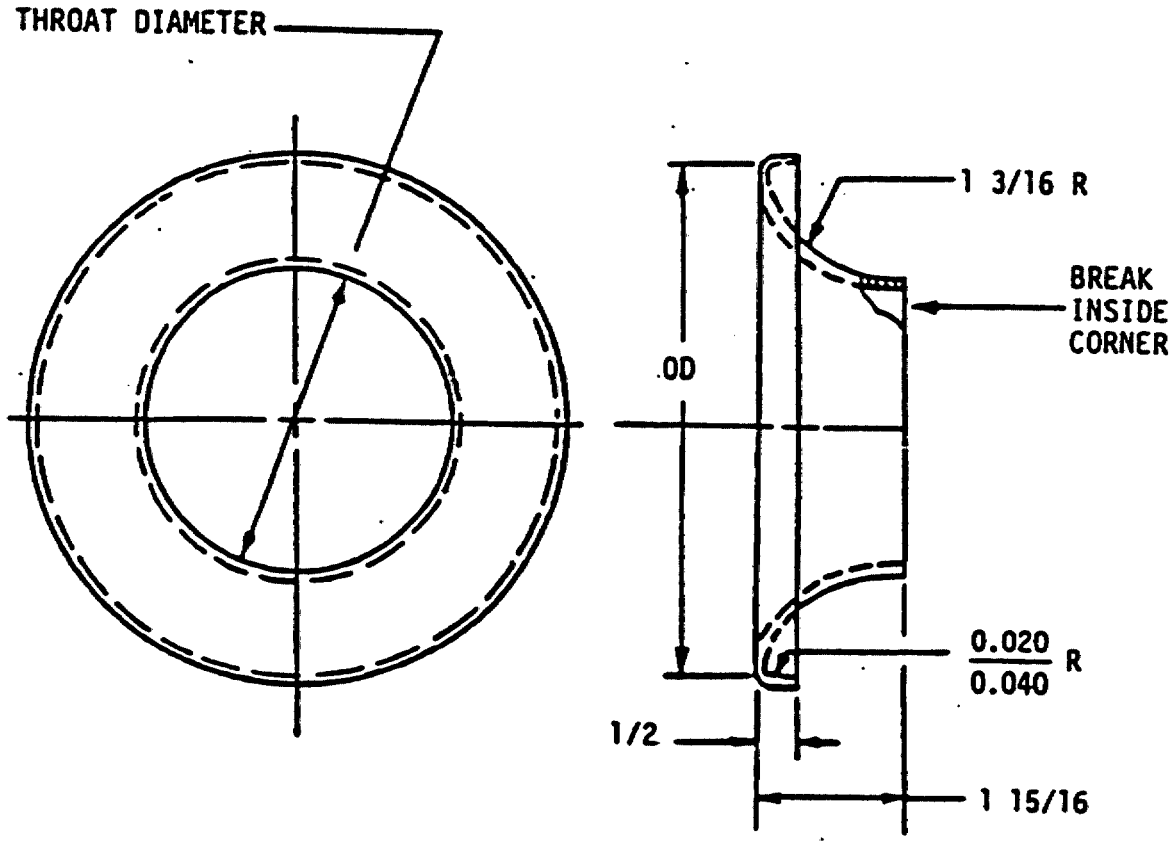
OUTLET PIEZOMETER TUBE

NOTES:

1. All dimensions are in inches.
2. Diameter (D), inside diameter (ID) and outside diameter (OD).
3. The OD shall be 7 1/2 for piezometer tubes 3 1/4 and larger ID; 4 for piezometer tubes smaller than 3 1/4 ID.

FIGURE 3. Piezometer tubes.

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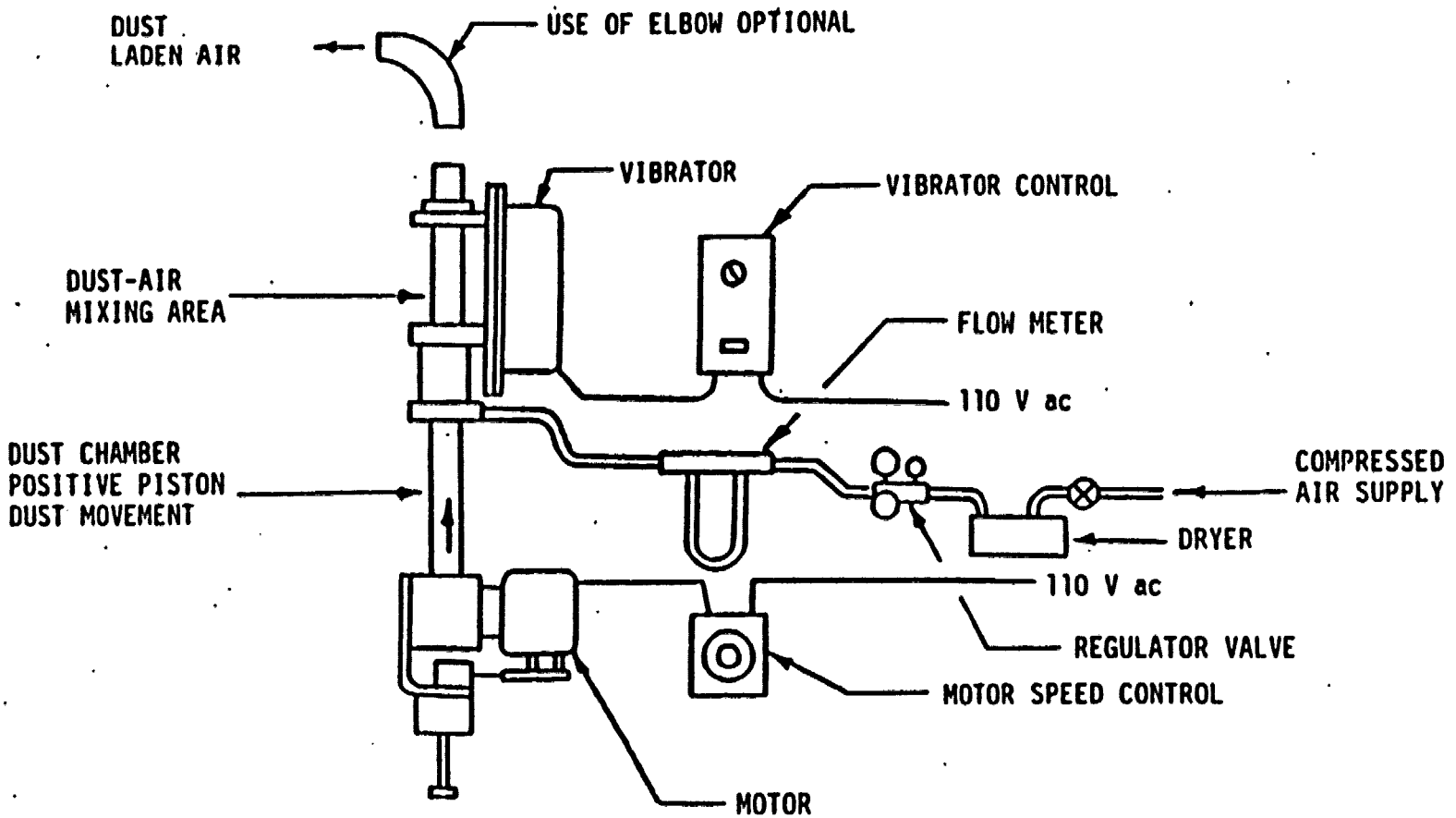
THROAT DIAMETER	CFM	THROAT DIAMETER	CFM
4 3/8	1,000	1 15/16	200
4 1/8	900	1 3/4	165
3 3/4	800	1 1/2	120
3 1/2	700	1 3/8	100
3 3/8	600	1 1/4	85
3 1/8	500	1	55
2 3/4	400	13/16	35
2 3/8	300	5/8	20
		7/16	10

NOTES:

1. Unless otherwise specified, all dimensions are in inches.
2. The OD shall be 7.500 + 0.010 or 4.000 + 0.010 diameter depending on piezometer tube diameter.
3. Radius (R).
4. Cubic feet per minute (CFM).

FIGURE 4. Ideal flow nozzle.

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NOTE: Volt (V), alternating current (ac).

FIGURE 5. Dust feeder.

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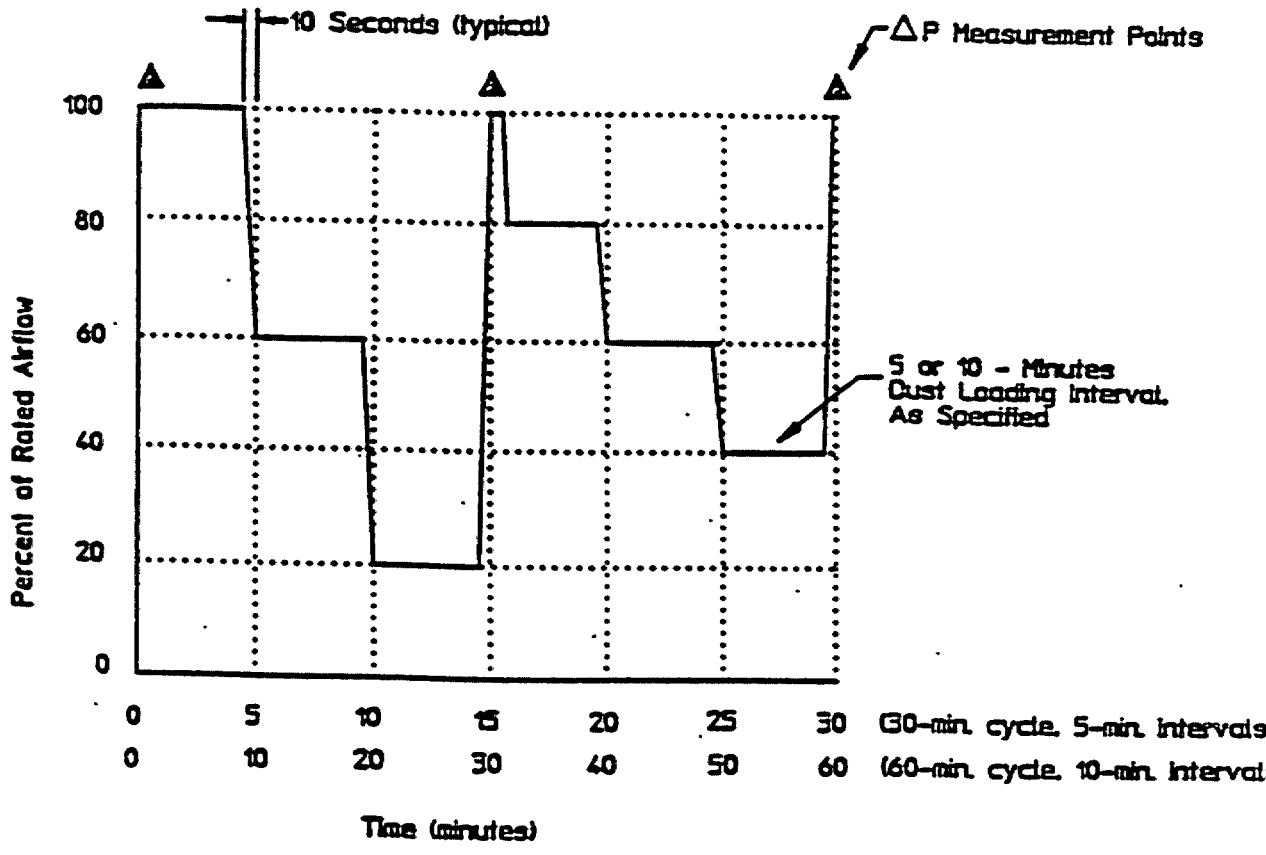
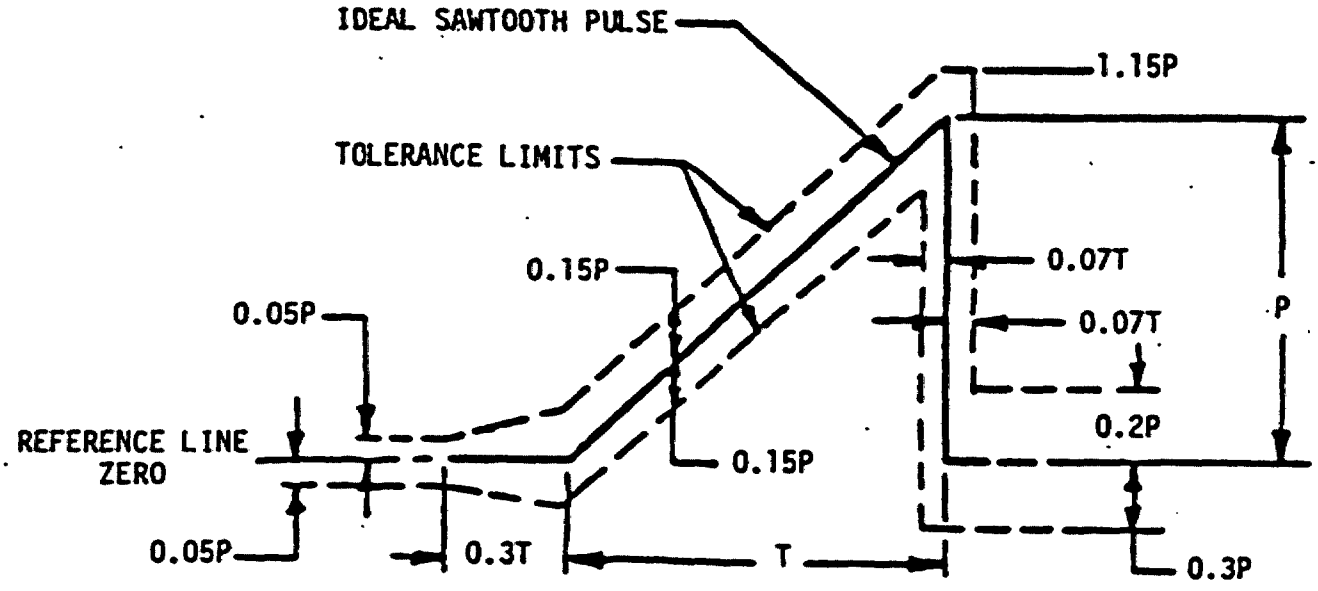


FIGURE 6. Variable Airflow Cycle.

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NOTES:

1. Peak value (P) = 40 g.
2. Nominal duration (T) = 11 ms.
3. The oscillogram shall include a time about 3T long with a pulse located approximately in the center. The peak acceleration magnitude of the sawtooth pulse is P and it's duration is T. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change (which may be obtained by integration of the acceleration pulse) shall be within the limits of Vi ± 0.1 Vi, where Vi is the velocity change associated with the ideal pulse which equals 0.5 PT. The integration to determine velocity change shall extend from 0.4T before the pulse to 0.1T after the pulse.

FIGURE 7. Terminal-peak sawtooth shock pulse configuration and tolerance limits.

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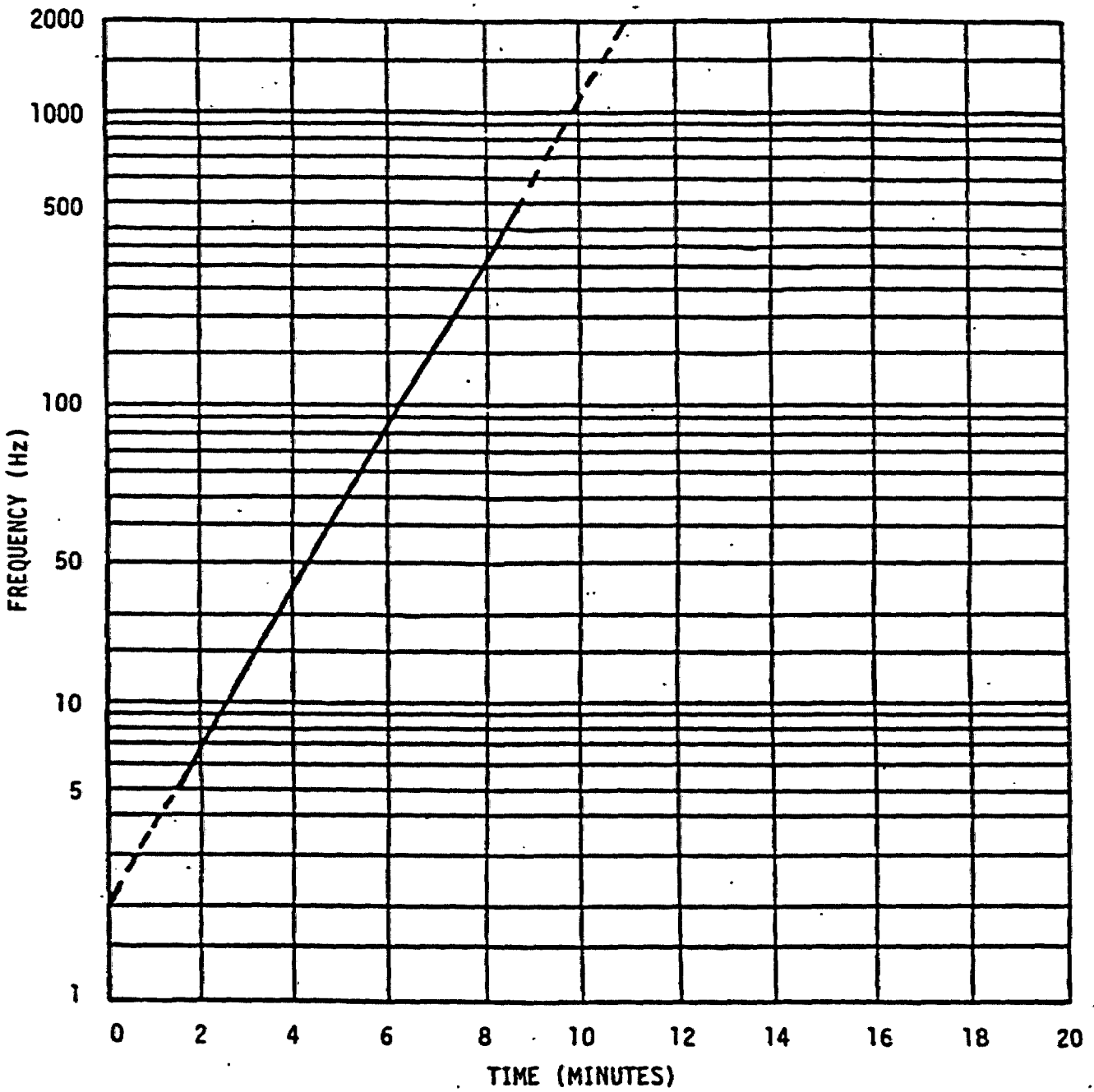


FIGURE 8. Logarithmic sweep.

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